

IR-8 Inorganic Acids and Derivatives

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IR-8.1 INTRODUCTION AND OVERVIEW

Certain inorganic and simple carbon-containing compounds are commonly given non-systematic or semi-systematic names containing the word 'acid'. Examples are boric acid or orthoboric acid, metaboric acid, phosphoric acid, diphosphoric acid, *cyclo*-triphosphoric acid, *catena*-triphosphoric acid, dithionous acid, peroxydisulfuric acid or peroxydisulfuric acid, *etc.* These names are unique in modern nomenclature in that, interpreted literally, they describe a particular chemical property of the compounds in question. Systematic names are otherwise based solely on composition and structure.

All such acids may also be given structure-based systematic names using principles already described in preceding chapters on substitutive and additive nomenclature, so in that respect the 'acid'-containing names are superfluous. Furthermore, many species which would be classified as acids based on their chemical properties are never named as such, *e.g.* aqua ions such as hexaaquaaluminium(3+), and hydrides and derivatives such as ammonium, hydrogen sulfide (sulfane), *etc.* The term 'acid' is thus not used consistently.

Based on these considerations, the use of the word 'acid' in any new name in inorganic nomenclature is discouraged. However, a number of the existing 'acid' names are so commonly used (sulfuric acid, perchloric acid, *etc.*) that it would be unrealistic to suggest replacing them altogether by systematic alternatives. Another reason to include them in the present recommendations is that the acids in question are used as parent structures in the nomenclature of certain organic (*i.e.* carbon-containing) derivatives so that the derivative names are directly or indirectly based on the names containing the word 'acid'. See examples below and Section IR-8.6.

The main purposes of this chapter are:

- (a) to show how the inorganic species commonly named as acids may be given systematic additive names (Section IR-8.3 and Tables IR-8.1 and IR-8.2);

- (b) to list the ‘acid’ names that are still acceptable due to common usage and/or because they are needed in organic nomenclature (see Tables IR-8.1 and IR-8.2).

In addition, Sections IR-8.4 and IR-8.5 deal with a further type of names, denoted here as *hydrogen names*. These names can be viewed as generalizations of common anion names such as ‘hydrogencarbonate’, but they are not necessary for naming completely specified molecular structures and can be regarded as a special topic.

Thus, this Chapter provides several acceptable names for many inorganic acids; it is left to practitioners to choose the name most suitable for a particular application. In the future, IUPAC aims to select *preferred* names for inorganic species, including the acids dealt with here, just as Ref. 1 does for organic species.

Finally, names which do not denote compounds of a definite composition, such as hydrochloric acid, stannic acid, tungstic acid, *etc.*, are outside the scope of the systematic nomenclature presented here. However, the chemical systems involved can always be discussed using systematic names such as hydrogen chloride, tin(IV) oxide, tungsten(VI) oxide, *etc.*

A few examples are given now in order to illustrate some of the general remarks above. In these examples, and in the remainder of this chapter, alternative formulae are sometimes provided for clarity in connection with the discussion of additive names. These are based on a perception of the structures in question as generalized coordination entities. For mononuclear entities, this means that the central atom symbol is listed first and then the ligand symbols in alphabetical order, as prescribed in Section IR-4.4.3.2.

Example:

1. phosphoric acid = H_3PO_4 or $[\text{PO}(\text{OH})_3]$

Based on the structure, the compound can be named substitutively (Chapter IR-6) as a derivative of the parent hydride λ^5 -phosphane (PH_5), leading to the name trihydroxy- λ^5 -phosphanone, or additively (Chapter IR-7) as trihydroxidooxidophosphorus.

As opposed to the two last names, the name phosphoric acid does not convey the structure, but does fit into a general pattern whereby the ending ‘ic’ denotes a higher or the highest possible oxidation state (compare nitric acid, sulfuric acid). Examples 2 and 3 show organic derivatives named on the basis of phosphoric acid as the parent.

Examples:

- | | |
|--------------------------------|-------------------------------|
| 2. $\text{PO}(\text{OMe})_3$ | trimethyl phosphate |
| 3. $\text{PO}(\text{NMe}_2)_3$ | hexamethylphosphoric triamide |

Each of these two compounds could also be named substitutively, on the basis of the above parent hydride, or additively but the names given here are preferred IUPAC names (see Section P-67.1 of Ref. 1).

Some organic derivative names still contain the word ‘acid’, as in the following derivatives of arsonic acid (H_2AsHO_3 or $[\text{AsHO}(\text{OH})_2]$).

Examples:

- | | |
|---------------------------------------|--|
| 4. $\text{PhAsO}(\text{OH})_2$ | phenylarsonic acid |
| 5. $\text{EtAsCl}(\text{OH})\text{S}$ | ethylarsonochloridothioic <i>O</i> -acid |

The name in Example 4 regards the compound as derived from arsonic acid, by substitution of a phenyl group for the hydrogen atom bound directly to arsenic. The name in Example 5, in addition to the hydrogen substitution, involves functional replacement nomenclature (Section IR-8.6).

Note that there is one general case where the word ‘acid’ may appear in a fully systematic name of an inorganic compound, namely when substitutive nomenclature is used and prescribes a suffix for the highest ranking substituent group which ends with the word ‘acid’.

Consider the polythionic acids, $\text{H}_2\text{S}_n\text{O}_6 = [(\text{HO})(\text{O})_2\text{SS}_{n-2}\text{S}(\text{O})_2(\text{OH})]$ ($n \geq 2$), which have the common names dithionic acid, trithionic acid, tetrathionic acid, *etc.* They may be named systematically using additive nomenclature, as shown in Table IR-8.1. For $n \geq 3$, they may also be named substitutively on the basis of the central (poly)sulfane skeleton, as exemplified below.

Examples:

- | | |
|---|--------------------------|
| 6. $\text{H}_2\text{S}_3\text{O}_6 = [(\text{HO})(\text{O})_2\text{SSS}(\text{O})_2(\text{OH})]$ | sulfanedisulfonic acid |
| 7. $\text{H}_2\text{S}_4\text{O}_6 = [(\text{HO})(\text{O})_2\text{SSSS}(\text{O})_2(\text{OH})]$ | disulfanedisulfonic acid |

IR-8.2 GENERAL PRINCIPLES FOR SYSTEMATIC NAMING OF ACIDS

Molecular compounds and ions commonly regarded as inorganic acids are treated no differently than other molecular species when constructing systematic names.

The most easily applied general principle for systematic naming is that of additive nomenclature, exemplified in Section IR-8.3. As mentioned in Section IR-8.1, substitutive nomenclature could also be generally applied. However, this is not further elaborated here.

Sections IR-8.4 and IR-8.5 describe hydrogen names, which are related to additive names and only needed in special cases.

The method called ‘acid nomenclature’ in Section I-9.6 of Ref. 2 is little used and not needed. Its use is therefore no longer recommended.

Table IR-8.1 *Acceptable common names and systematic (additive) names for oxoacid and related structures*

This Table includes compounds containing oxygen and hydrogen and at least one other element and with at least one OH group; certain isomers; and examples of corresponding partially and fully dehydrated species. Formulae are given in the classical oxoacid format with the 'acid' (oxygen-bound) hydrogens listed first, followed by the central atom(s), then the hydrogen atoms bound directly to the central atom, and then the oxygen atoms (*e.g.* $\text{H}_2\text{P}_2\text{H}_2\text{O}_5$), *except* for chain compounds such as *e.g.* HOCN . In most cases formulae are also written as for coordination entities, assembled according to the principles of Chapter IR-7 (*e.g.* the Table gives ' H_2BO_3 ' as $[\text{B}(\text{OH})_3]$ and ' H_2SO_4 ' as $[\text{SO}_2(\text{OH})_2]$ '). More names of oxoanions are given in Table IX.

Note that Section P-42 of Ref. 1 lists a great many inorganic oxoacid species for use as parent structures in the naming of organic derivatives. (See the discussion in Section IR-8.1.) Most of those species, but not all, are included here. In particular, several dinuclear and polynuclear acids are not explicitly included.

<i>Formula</i>	<i>Acceptable common name (unless otherwise stated)</i>	<i>Systematic additive name(s)</i>
$\text{H}_3\text{BO}_3 = [\text{B}(\text{OH})_3]$	boric acid ^a	trihydroxidoboron
$\text{H}_2\text{BO}_3^- = [\text{BO}(\text{OH})_2]^-$	dihydrogenborate	dihydroxidooxidoborate(1-)
$\text{HBO}_3^{2-} = [\text{BO}_2(\text{OH})]^{2-}$	hydrogenborate	hydroxidodioxidoborate(2-)
$[\text{BO}_3]^{3-}$	borate	trioxidoborate(3-)
$(\text{HBO}_2)_n = \text{-(B}(\text{OH})\text{O)-}_n$	metaboric acid	<i>catena</i> -poly[hydroxidoboron- μ -oxido]
$(\text{BO}_2^-)_n = \text{-(OBO)}_n^{n-}$	metaborate	<i>catena</i> -poly[(oxidoborate- μ -oxido)(1-)]
$\text{H}_2\text{BHO}_2 = [\text{BH}(\text{OH})_2]$	boronic acid	hydridodihydroxidoboron
$\text{HBH}_2\text{O} = [\text{BH}_2(\text{OH})]$	borinic acid	dihydridohydroxidoboron
$\text{H}_2\text{CO}_3 = [\text{CO}(\text{OH})_2]$	carbonic acid	dihydroxidooxidocarbon
$\text{HCO}_3^- = [\text{CO}_2(\text{OH})]^-$	hydrogencarbonate	hydroxidodioxidocarbonate(1-)
$[\text{CO}_3]^{2-}$	carbonate	trioxidocarbonate(2-)
$\text{HOCN} = [\text{C}(\text{N})\text{OH}]$	cyanic acid	hydroxidonitridocarbon
$\text{HNCO} = [\text{C}(\text{NH})\text{O}]$	isocyanic acid	azanediidioxidocarbon, (hydridonitrate)oxidocarbon
$\text{OCN}^- = [\text{C}(\text{N})\text{O}]^-$	cyanate	nitridooxidocarbonate(1-)
$\text{HONC} = [\text{N}(\text{C})\text{OH}]$	^b	carbidohydroxidodinitrogen
$\text{HCNO} = [\text{N}(\text{C})\text{O}]$	^b	(hydridocarbonate)oxidodinitrogen
$\text{ONC}^- = [\text{N}(\text{C})\text{O}]^-$	^b	carbidooxidonitrate(1-)
$\text{H}_4\text{SiO}_4 = [\text{Si}(\text{OH})_4]$	silicic acid ^a	tetrahydroxidossilicon
$[\text{SiO}_4]^{4-}$	silicate	tetraoxidosilicate(4-)
$(\text{H}_2\text{SiO}_3)_n = \text{-(Si}(\text{OH})_2\text{O)-}_n$	metasilicic acid	<i>catena</i> -poly[dihydroxidossilicon- μ -oxido]

(continued)

Table IR-8.1 Continued

Formula	Acceptable common name (unless otherwise stated)	Systematic additive name(s)
$\{ \text{SiO}_3 \}_n^{2n-}$	metasilicate	catena-poly[(dioxidosilicate- μ -oxido)(1-)]
$\text{H}_6\text{Si}_2\text{O}_7 = [(\text{HO})_3\text{SiOSi}(\text{OH})_3]$	disilicic acid ^e	μ -oxido-bis(trihydroxidosilicon)
$[\text{Si}_2\text{O}_7]^{6-} = [\text{O}_3\text{SiOSiO}_3]^{6-}$	disilicate	μ -oxido-bis(trioxidosilicate)(6-)
$\text{H}_2\text{NO}_3^+ = [\text{NO}(\text{OH})_2]^+$	^d	dihydroxidooxidonitrogen(1+), dihydrogen(trioxidonitrate)(1+)
$\text{HNO}_3 = [\text{NO}_2(\text{OH})]$	nitric acid	hydroxidoxidonitrogen
$[\text{NO}_3]^-$	nitrate	trioxidonitrate(1-)
$\text{H}_2\text{NHO} = [\text{NH}_2\text{OH}]$	hydroxylamine ^e	dihydrodihydroxidonitrogen
$\text{H}_2\text{NHO}_3 = [\text{NHO}(\text{OH})_2]$	azonic acid	hydridodihydroxidooxidonitrogen
$\text{HNO}_2 = [\text{NO}(\text{OH})]$	nitrous acid	hydroxidooxidonitrogen
$[\text{NO}_2]^-$	nitrite	dioxidonitrate(1-)
$\text{HNH}_2\text{O}_2 = [\text{NH}_2\text{O}(\text{OH})]$	azinic acid	dihydrodihydroxidooxidonitrogen
$\text{H}_2\text{N}_2\text{O}_2 = [\text{HON}=\text{NOH}]$	diazenediol ^f	bis(hydroxidonitrogen)($N-N$), or 1,4-dihydro-2,3-diazy-[4]catena
$\text{HN}_2\text{O}_2^- = [\text{HON}=\text{NO}]^-$	2-hydroxydiazene-1-olate ^f	hydroxido-1 κ O-oxido-2 κ O-dinitrate($N-N$)(1-), or 1-hydro-2,3-diazy-1,4-dioxy-[4]catenate(1-)
$[\text{N}_2\text{O}_2]^{2-} = [\text{ON}=\text{NO}]^{2-}$	diazenediolate ^f	bis(oxidonitrate)($N-N$)(2-), or 2,3-diazy-[4]catenate(2-)
$\text{H}_3\text{PO}_4 = [\text{PO}(\text{OH})_3]$	phosphoric acid ^a	trihydroxidooxiphosphorus
$\text{H}_2\text{PO}_4^- = [\text{PO}_2(\text{OH})_2]^-$	dihydrogenphosphate	dihydroxidoxidoxiphosphate(1-)
$\text{HPO}_4^{2-} = [\text{PO}_3(\text{OH})]^{2-}$	hydrogenphosphate	hydroxidotrioxiphosphate(2-)
$[\text{PO}_4]^{3-}$	phosphate	tetraoxiphosphate(3-)
$\text{H}_2\text{PHO}_3 = [\text{PHO}(\text{OH})_2]$	phosphonic acid ^g	hydridodihydroxidoxidoxiphosphorus
$[\text{PHO}_2(\text{OH})]^-$	hydrogenphosphonate	hydridohydroxidoxidoxiphosphate(1-)
$[\text{PHO}_3]^{2-}$	phosphonate	hydridotrioxiphosphate(2-)
$\text{H}_3\text{PO}_3 = [\text{P}(\text{OH})_3]$	phosphorous acid ^g	trihydroxidoxiphosphorus
$\text{H}_2\text{PO}_3^- = [\text{PO}(\text{OH})_2]^-$	dihydrogenphosphite	dihydroxidoxidoxiphosphate(1-)
$\text{HPO}_3^{2-} = [\text{PO}_2(\text{OH})]^{2-}$	hydrogenphosphite	hydroxidoxidoxiphosphate(2-)
$[\text{PO}_3]^{3-}$	phosphite	trioxiphosphate(3-)
$\text{HPO}_2 = [\text{P}(\text{O})\text{OH}]$	hydroxyphosphanone ^h	hydroxidoxidoxiphosphorus
$\text{HPO}_2 = [\text{P}(\text{H})\text{O}_2]$	λ^5 -phosphanedione ^h	hydridodioxidoxiphosphorus
$\text{H}_2\text{PHO}_2 = [\text{PH}(\text{OH})_2]$	phosphonous acid	hydridodihydroxidoxiphosphorus

$\text{HPH}_2\text{O}_2 = [\text{PH}_2\text{O}(\text{OH})]$	phosphinic acid	dihydrodihydrodioxido phosphorus
$\text{HPH}_2\text{O} = [\text{PH}_2(\text{OH})]$	phosphinous acid	dihydrodihydrodioxido phosphorus
$\text{H}_4\text{P}_2\text{O}_7 = [(\text{HO})_2\text{P}(\text{O})\text{OP}(\text{O})(\text{OH})_2]$	diphosphoric acid ^e	μ -oxido-bis(dihydrodioxido phosphorus)
$(\text{HPO}_3)_n = 4\text{P}(\text{O})(\text{OH})\text{O}_n$	metaphosphoric acid	<i>catena</i> -poly[hydrodioxido phosphorus- μ -oxido]
$\text{H}_4\text{P}_2\text{O}_6 = [(\text{HO})_2\text{P}(\text{O})\text{P}(\text{O})(\text{OH})_2]$	hypodiphosphoric acid	bis(dihydrodioxido phosphorus)(<i>P</i> — <i>P</i>)
$\text{H}_2\text{P}_2\text{H}_2\text{O}_5 = [(\text{HO})\text{P}(\text{H})(\text{O})\text{OP}(\text{H})(\text{O})(\text{OH})]$	diphosphonic acid	μ -oxido-bis(hydrodihydrodioxido phosphorus)
$\text{P}_2\text{H}_2\text{O}_5^{2-} = [\text{O}_2\text{P}(\text{H})\text{OP}(\text{H})(\text{O})]^{2-}$	diphosphonate	μ -oxido-bis(hydrodihydrodioxido phosphate)(2—)
$\text{H}_3\text{P}_3\text{O}_9$	<i>cyclo</i> -triphosphoric acid	tri- μ -oxido-tris(hydrodioxido phosphorus), or
		2,4,6-trihydroxido-2,4,6-trioxido-1,3,5-trioxy-2,4,6-triphosphoryl-[6]cycle
$\text{H}_3\text{P}_3\text{O}_{10}$	<i>catena</i> -triphosphoric acid	pentahydroxido-1 κ^2 O,2 κ O,3 κ^2 O-di- μ -oxido-trioxido-1 κ O,2 κ O,3 κ O-triphosphorus, or μ -(hydrodihydrodioxido phosphate-1 κ O,2 κ O)-bis(dihydrodioxido phosphorus),
		or 1,7-dihydroxido-2,4,6-trihydroxido-2,4,6-trioxido-1,3,5,7-tetraoxy-2,4,6-triphosphoryl-[7] <i>catena</i>
$\text{H}_3\text{AsO}_4 = [\text{AsO}(\text{OH})_3]$	arsenic acid, arsonic acid ⁱ	trihydrodioxidoarsenic
$\text{H}_3\text{AsO}_3 = [\text{As}(\text{OH})_3]$	arsenous acid, arsonous acid ⁱ	trihydroxidoarsenic
$\text{H}_2\text{AsHO}_3 = [\text{AsHO}(\text{OH})_2]$	arsonic acid	hydrodihydrodioxidoarsenic
$\text{H}_2\text{AsHO}_2 = [\text{AsH}(\text{OH})_2]$	arsonous acid	hydrodihydrodioxidoarsenic
$\text{HAS}_2\text{O}_2 = [\text{AsH}_2\text{O}(\text{OH})]$	arsinic acid	dihydrodihydrodioxidoarsenic
$\text{HAS}_2\text{O} = [\text{AsH}_2(\text{OH})]$	arsinous acid	dihydrodihydrodioxidoarsenic
$\text{H}_3\text{SbO}_4 = [\text{SbO}(\text{OH})_3]$	antimonic acid, stiboric acid ⁱ	trihydrodioxidoantimony
$\text{H}_3\text{SbO}_3 = [\text{Sb}(\text{OH})_3]$	antimonous acid, stiborous acid ⁱ	trihydroxidoantimony
$\text{H}_2\text{SbHO}_3 = [\text{SbHO}(\text{OH})_2]$	stibonic acid	hydrodihydrodioxidoantimony
$\text{H}_2\text{SbHO}_2 = [\text{SbH}(\text{OH})_2]$	stibonous acid	hydrodihydrodioxidoantimony
$\text{HSb}_2\text{O}_2 = [\text{SbH}_2\text{O}(\text{OH})]$	stibinic acid	dihydrodihydrodioxidoantimony
$\text{HSb}_2\text{O} = [\text{SbH}_2(\text{OH})]$	stibinous acid	dihydrodihydrodioxidoantimony
$\text{H}_3\text{SO}_4^+ = [\text{SO}(\text{OH})_3]^+$	^d	trihydrodioxidosulfur(1+), trihydrogen(tetraoxidosulfate)(1+)
$\text{H}_2\text{SO}_4 = [\text{SO}_2(\text{OH})_2]$	sulfuric acid	dihydrodioxidosulfur
$\text{HSO}_4^- = [\text{SO}_3(\text{OH})]^-$	hydrogensulfate	hydroxido trioxidosulfate(1—)
$[\text{SO}_4]^{2-}$	sulfate	tetraoxidosulfate(2—)
$\text{HSO}_3 = [\text{SHO}_2(\text{OH})]$	sulfonic acid ^j	hydrodihydrodioxidosulfur
$\text{H}_2\text{SO}_3 = [\text{SO}(\text{OH})_2]$	sulfurous acid	dihydrodioxidosulfur

(continued)

Table IR-8.1 Continued

Formula	Acceptable common name (unless otherwise stated)	Systematic additive name(s)
$\text{HSO}_3^- = [\text{SO}_2(\text{OH})]^-$	hydrogensulfite	hydroxidodioxidosulfate(1-)
$[\text{SO}_3]^{2-}$	sulfite	trioxidosulfate(2-)
$\text{HSO}_2 = [\text{SHO}(\text{OH})]$	sulfinic acid ^j	hydridohydroxidodioxosulfur
$\text{H}_2\text{SO}_2 = [\text{S}(\text{OH})_2]$	sulfanedio ^k	dihydroxidodioxosulfur
$[\text{SO}_2]^{2-}$	sulfanediolate ^k	dioxidosulfate(2-)
$\text{HSOH} = [\text{SH}(\text{OH})]$	sulfanol ^k	hydridohydroxidodioxosulfur
$\text{HSO}^- = [\text{SHO}]^-$	sulfanolate ^k	hydridodioxidosulfate(1-)
$\text{H}_2\text{S}_2\text{O}_7 = [(\text{HO})\text{S}(\text{O})_2\text{OS}(\text{O})_2(\text{OH})]$	disulfuric acid ^e	μ -oxido-bis(hydroxidodioxidosulfur)
$[\text{S}_2\text{O}_7]^{2-} = [(\text{O})_3\text{SOS}(\text{O})_3]^{2-}$	disulfate	μ -oxido-bis(trioxidosulfate)(2-)
$\text{H}_2\text{S}_2\text{O}_6 = [(\text{HO})(\text{O})_2\text{SS}(\text{O})_2(\text{OH})]$	dithionic acid ^{c,l}	bis(hydroxidodioxidosulfur)(S—S), or 1,4-dihydrido-2,2,3,3-tetraoxido-1,4-dioxy-2,3-disulfo-[4]catena
$[\text{S}_2\text{O}_6]^{2-} = [\text{O}_3\text{SSO}_3]^{2-}$	dithionate	bis(trioxidosulfate)(S—S)(2-), or 2,2,3,3-tetraoxido-1,4-dioxy-2,3-disulfo-[4]catenate(2-)
$\text{H}_2\text{S}_3\text{O}_6 = [(\text{HO})(\text{O})_2\text{SSS}(\text{O})_2(\text{OH})]$	trithionic acid ^{c,m}	or 1,5-dihydrido-2,2,4,4-tetraoxido-1,5-dioxy-2,3,4-trisulfo-[5]catena
$\text{H}_2\text{S}_4\text{O}_6 = [(\text{HO})(\text{O})_2\text{SSSS}(\text{O})_2(\text{OH})]$	tetrathionic acid ^{c,m}	1,6-dihydrido-2,2,5,5-tetraoxido-1,6-dioxy-2,3,4,5-tetrasulfo-[6]catena
$\text{H}_2\text{S}_2\text{O}_5 = [(\text{HO})(\text{O})_2\text{SS}(\text{O})\text{OH}]$	disulfurous acid ⁿ	dihydroxido-1 κ O,2 κ O-trioxido-1 κ^2 O,2 κ O-disulfo(S—S)(2-)
$[\text{S}_2\text{O}_5]^{2-} = [(\text{O})_2\text{SS}(\text{O})\text{O}]^{2-}$	disulfite ⁿ	pentaoxido-1 κ^3 O,2 κ^2 O-disulfo(S—S), bis(hydroxidodioxidosulfur)(S—S),
$\text{H}_2\text{S}_2\text{O}_4 = [(\text{HO})(\text{O})\text{SS}(\text{O})(\text{OH})]$	dithionous acid ^{c,l}	or 1,4-dihydrido-2,3-dioxido-1,4-dioxy-2,3-disulfo-[4]catena
$[\text{S}_2\text{O}_4]^{2-} = [\text{O}_2\text{SSO}_2]^{2-}$	dithionite	bis(dioxidosulfate)(S—S)(2-), or 2,3-dioxido-1,4-dioxy-2,3-disulfo-[4] catenate(2-)
$\text{H}_2\text{SeO}_4 = [\text{SeO}_2(\text{OH})_2]$	selenic acid	dihydroxidodioxidoselenium
$[\text{SeO}_4]^{2-}$	selenate	tetraoxidoselenate(2-)
$\text{H}_2\text{SeO}_3 = [\text{SeHO}_2(\text{OH})]$	selenonic acid ^{j,o}	hydridohydroxidodioxidoselenium
$\text{H}_2\text{SeO}_3 = [\text{SeO}(\text{OH})_2]$	selenous acid ^o	dihydroxidodioxidoselenium
$[\text{SeO}_3]^{2-}$	selenite	trioxidoselenate(2-)
$\text{HSeHO}_2 = [\text{SeHO}(\text{OH})]$	seleninic acid ^j	hydridohydroxidodioxidoselenium
$\text{H}_6\text{TeO}_6 = [\text{Te}(\text{OH})_6]$	orthotelluric acid ^a	hexahydroxidotellurium
$[\text{TeO}_6]^{6-}$	orthotellurate ^a	hexaoxidotellurate(6-)

$\text{H}_2\text{TeO}_4 = [\text{TeO}_2(\text{OH})_2]$	telluric acid ^a	dihydrodioxidotellurium
$[\text{TeO}_4]^{2-}$	tellurate ^a	tetraoxidotellurate(2-)
$\text{H}_2\text{TeO}_3 = [\text{TeO}(\text{OH})_2]$	tellurous acid	dihydrodioxidotellurium
$\text{HTeHO}_3 = [\text{TeHO}_2(\text{OH})]$	telluronic acid ^b	hydrodihydrodioxidotellurium
$\text{HTeHO}_2 = [\text{TeHO}(\text{OH})]$	tellurinic acid ^c	hydridohydrodioxidotellurium
$\text{HClO}_4 = [\text{ClO}_3(\text{OH})]$	perchloric acid	hydroxidotrioxidochlorine
$[\text{ClO}_4]^-$	perchlorate	tetraoxidochlorate(1-)
$\text{HClO}_3 = [\text{ClO}_2(\text{OH})]$	chloric acid	hydroxidodioxidochlorine
$[\text{ClO}_3]^-$	chlorate	trioxidochlorate(1-)
$\text{HClO}_2 = [\text{ClO}(\text{OH})]$	chlorous acid	hydroxidooxidochlorine
$[\text{ClO}_2]^-$	chlorite	dioxidochlorate(1-)
$\text{HClO} = [\text{O}(\text{H})\text{Cl}]$	hypochlorous acid	chloridohydrodooxygen
$[\text{OCl}]^-$	hypochlorite	chloridooxygenate(1-)
$\text{HBrO}_4 = [\text{BrO}_3(\text{OH})]$	perbromic acid	hydroxidotrioxidobromine
$[\text{BrO}_4]^-$	perbromate	tetraoxidobromate(1-)
$\text{HBrO}_3 = [\text{BrO}_2(\text{OH})]$	bromic acid	hydroxidodioxidobromine
$[\text{BrO}_3]^-$	bromate	trioxidobromate(1-)
$\text{HBrO}_2 = [\text{BrO}(\text{OH})]$	bromous acid	hydroxidooxidobromine
$[\text{BrO}_2]^-$	bromite	dioxidobromate(1-)
$\text{HBrO} = [\text{O}(\text{H})\text{Br}]$	hypobromous acid	bromidohydrodooxygen
$[\text{OBr}]^-$	hypobromite	bromidooxygenate(1-)
$\text{H}_5\text{IO}_6 = [\text{IO}(\text{OH})_5]$	orthoperiodic acid ^a	pentahydrodioxidoiodine
$[\text{IO}_6]^{5-}$	orthoperiodate ^a	hexaoxidoiodate(5-)
$\text{HIO}_4 = [\text{IO}_3(\text{OH})]$	periodic acid ^a	hydroxidotrioxidoiodine
$[\text{IO}_4]^-$	periodate ^a	tetraoxidoiodate(1-)
$\text{HIO}_3 = [\text{IO}_2(\text{OH})]$	iodic acid	hydroxidodioxidoiodine
$[\text{IO}_3]^-$	iodate	trioxidoiodate(1-)
$\text{HIO}_2 = [\text{IO}(\text{OH})]$	iodous acid	hydroxidooxidoiodine
$[\text{IO}_2]^-$	iodite	dioxidoiodate(1-)
$\text{HIO} = [\text{O}(\text{H})\text{I}]$	hypoiodous acid	hydridoiodidooxygen
$[\text{OI}]^-$	hypoiodite	iodidooxygenate(1-)

(continued)

- ^a The prefix 'ortho' has not been used consistently in the past (including in Chapter I-9 of Ref. 2). Here, it has been removed in the cases of boric acid, silicic acid and phosphoric acid where there is no ambiguity in the names without 'ortho'. The only cases where 'ortho' distinguishes between two different compounds are the telluric and periodic acids (and corresponding anions).
- ^b The names fulminic acid and isofulminic acid have been used inconsistently in the past. The compound originally named fulminic acid is HCNO, which is not an oxoacid, while the esters usually called fulminates in organic chemistry are RONC, corresponding to the oxoacid HONC. The additive names in the right hand column specify the structures unambiguously. The preferred organic names are formonitrile oxide for HCNO and λ^2 -methylidenehydroxylamine for HONC. (See Section P-61.9 of Ref. 1. See also Table IX under entries CHNO and CNO).
- ^c The oligomeric series can be continued, *e.g.* diphosphoric acid, triphosphoric acid, *etc.*; dithionic acid, tetrathionic acid, *etc.*; dithionous, trithionous, *etc.*
- ^d The names nitric acidium, sulfuric acidium, *etc.* for the hydronated acids represent a hybrid of several nomenclatures and are difficult to translate into certain languages. They are no longer acceptable.
- ^e The substitutive name would be azanol. However, for preferred names for certain organic derivatives, NH_2OH itself is regarded as a parent with the name hydroxylamine. See Ref. 1, Section P-68.3.
- ^f These are systematic substitutive names. The traditional names hyponitrous acid and hyponitrite are not acceptable; the systematics otherwise adhered to for use of the prefix 'hypo' would have prescribed hypodinitrous and hypodinitrite.
- ^g The name phosphorous acid and the formula H_3PO_3 have been used in the literature for both $[\text{P}(\text{OH})_3]$ and $[\text{PHO}(\text{OH})_2]$. The present choice of names for these two species is in accord with the parent names given in Sections P-42.3 and P-42.4 of Ref. 1.
- ^h These are substitutive names. No 'acid' names are commonly used for the two isomers of HPO_2 .
- ⁱ The names arsorice, arsorous, stiboric and stiborous are included because they are used as parent names in Section P-42.4 of Ref. 1.
- ^j Caution is needed if using the names sulfonic acid, sulfenic acid, selenonic acid, *etc.* for these compounds. Substitutive nomenclature prescribes using substitution into parent hydrides rather than into the acids when naming corresponding functional derivatives, *e.g.* trisulfanedisulfonic acid (*not* trisulfanediyl...), see footnote m; methaneselenenic acid (*not* methyl...); *etc.* Note that the substituent groups 'sulfonyl', 'sulfinyl', *etc.*, are $-\text{S}(\text{O})_2-$, $-\text{S}(\text{O})-$, *etc.*, *not* $\text{HS}(\text{O})<$, $\text{HS}(\text{O})-$, *etc.*
- ^k These are systematic substitutive names. Names based on the traditional names sulfoxylic acid for $\text{S}(\text{OH})_2$ and sulfenic acid for HSOH , and indeed these names themselves, are no longer acceptable.
- ^l Systematic use of the prefix 'hypo' would give the names hypodisulfuric acid for dithionic acid and hypodisulfurous acid for dithionous acid.
- ^m The homologues trithionic acid, tetrathionic acid, *etc.*, may be alternatively named by substitutive nomenclature as sulfanedisulfonic acid, disulfanedisulfonic acid, *etc.*
- ⁿ This common name presents a problem because the unsymmetrical structure is not the structure which would otherwise be associated with the 'diacid' construction (disulfurous acid would systematically be $[\text{HO}(\text{O})\text{SOS}(\text{O})\text{OH}]$). The use of an additive name eliminates this potential confusion, but the problem with the use of disulfurous acid as a parent name persists in the naming of organic derivatives.
- ^o The formula H_2SeO_3 has been used in the literature for both selenonic acid and selenous acid. The present choice of names for the two structures shown is in accord with the parent names given in Sections P-42.1 and P-42.4 of Ref. 1.

IR-8.3

ADDITIVE NAMES

Molecules or ions that can *formally* be regarded as mononuclear coordination entities may be named additively, applying the rules described in Chapter IR-7.

Examples:

1. $\text{H}_3\text{SO}_4^+ = [\text{SO}(\text{OH})_3]^+$ trihydroxidooxidosulfur(1+)
2. $\text{H}_2\text{SO}_4 = [\text{SO}_2(\text{OH})_2]$ dihydroxidodioxidosulfur
3. $\text{HSO}_4^- = [\text{SO}_3(\text{OH})]^-$ hydroxidotrioxidosulfate(1-)

Structures which can be regarded as oligonuclear coordination entities may be named as such (Section IR-7.3) or may be named using the system for inorganic chains and rings (Section IR-7.4).

In principle, the choice of method in the latter case is arbitrary. However, the machinery of coordination compound nomenclature was developed to enable the handling of complex structures involving polyatomic, and particularly polydentate, ligands and sometimes multiply bridging ligands. Furthermore, the separation into ligands and central atoms, obvious in compounds most usually classified as coordination compounds, may be less obvious in the polyoxoacids. Thus, additive nomenclature of the coordination type tends to be more intricate than necessary when naming polyoxoacids forming relatively simple chains and rings. Here the chains and rings system is easily applied, and the names so derived are easy to decipher. However, this system can lead to long names with many locants.

Both types of additive names are exemplified below for oligonuclear systems.

Examples:

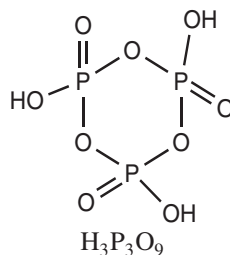
4. The compound commonly named diphosphoric acid, $\text{H}_4\text{P}_2\text{O}_7 = [(\text{HO})_2\text{P}(\text{O})\text{OP}(\text{O})(\text{OH})_2]$, is named according to the coordination-type additive nomenclature as:

μ -oxido-bis[dihydroxidooxidophosphorus]

or as a five-membered chain with ligands:

1,5-dihydro-2,4-dihydroxido-2,4-dioxido-1,3,5-trioxy-2,4-diphosphy-[5]catena

5. The compound commonly named *cyclo*-triphosphoric acid:

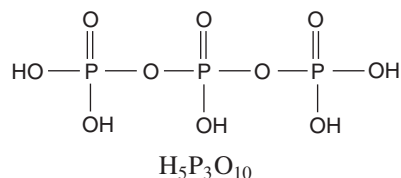


may be named according to coordination-type additive nomenclature as:

tri- μ -oxido-tris(hydroxidooxidophosphorus),

or as a six-membered ring with ligands:

2,4,6-trihydroxido-2,4,6-trioxido-1,3,5-trioxy-2,4,6-triphosphy-[6]cycle

6. The related compound, *catena*-triphosphoric acid

may be named as a trinuclear coordination entity:

pentahydroxido-1 κ^2 O,2 κ^2 O,3 κ O-di- μ -oxido-1:3 κ^2 O;2:3 κ^2 O-trioxido-1 κ O,2 κ O,3 κ O-triphosphorus,

or as a symmetrical dinuclear coordination entity with a bridging phosphate ligand:

μ -(hydroxidotrioxido-1 κ O,2 κ O'-phosphato)-bis(dihydroxidooxidophosphorus),

or as a mononuclear coordination entity with two phosphate ligands:

bis(dihydroxidodioxidophosphato)hydroxidooxidophosphorus,

or as a seven-membered chain with ligands:

1,7-dihydrido-2,4,6-trihydroxido-2,4,6-trioxido-1,3,5,7-tetraoxy-2,4,6-triphospho-[7]catena.

All inorganic oxoacids for which a common name containing the word 'acid' is still acceptable according to the present recommendations are listed in Table IR-8.1 together with additive names to illustrate how systematic names may be given.

Several names omitted from Ref. 2, *e.g.* selenic acid and hypobromous acid, are reinstated because they are unambiguous and remain in common use (including their use as parent names in functional replacement nomenclature, see Section IR-8.6).

Table IR-8.1 also includes anions derived from the neutral oxoacids by successive dehydration. Many of these anions also have common names that are still acceptable, in some cases in spite of the fact that they are based on nomenclature principles that are now otherwise abandoned (*e.g.* nitrate/nitrite and perchlorate/chlorate/chlorite/hypochlorite). For names involving the prefix 'hydrogen', see Sections IR-8.4 and IR-8.5.

It is important to note that the presence of a species in Table IR-8.1 does not imply that it has been described in the literature or that there has been a need to name it in the past. Several names are included only for completeness and to make parent names available for naming organic derivatives.

IR-8.4

HYDROGEN NAMES

An alternative nomenclature for hydrogen-containing compounds and ions is described here. The word 'hydrogen', with a multiplicative prefix if relevant, is joined (with no space) to an anion name formed by additive nomenclature and placed within appropriate enclosing marks (see Section IR-2.2). This construction is followed (again with no space) by a charge number indicating the total charge of the species or structural unit being named (except for neutral species/units).

Hydrogen names are useful when the connectivity (the positions of attachment of the hydrons) in a hydron-containing compound or ion is unknown or not specified (*i.e.* when which of two or more tautomers is not specified, or when one does not wish to specify a complex connectivity, such as in network compounds).

Some of the following examples are discussed in detail below.

Examples:

1. $\text{H}_2\text{P}_2\text{O}_7^{2-}$
dihydrogen(diphosphate), or
dihydrogen[μ -oxidobis(trioxidophosphate)](2-)
2. $\text{H}_2\text{B}_2(\text{O}_2)_2(\text{OH})_4$
dihydrogen(tetrahydroxidodi- μ -peroxido-diborate)
3. $\text{H}_2\text{Mo}_6\text{O}_{19} = \text{H}_2[\text{Mo}_6\text{O}_{19}]$
dihydrogen(nonadecaoxidohexamolybdate)
4. $\text{H}_4[\text{SiW}_{12}\text{O}_{40}] = \text{H}_4[\text{W}_{12}\text{O}_{36}(\text{SiO}_4)]$
tetrahydrogen[(tetracontaoxidosiliconododecatungsten)ate], or
tetrahydrogen[hexatriacontaoxido(tetraoxidosilicato)dodecatungstate], or
tetrahydrogen(silicododecatungstate)
5. $\text{H}_4[\text{PMo}_{12}\text{O}_{40}] = \text{H}_4[\text{Mo}_{12}\text{O}_{36}(\text{PO}_4)]$
tetrahydrogen[tetracontaoxido(phosphorusdodecamolybdenum)ate], or
tetrahydrogen[hexatriacontaoxido(tetraoxidophosphato)dodecamolybdate], or
tetrahydrogen(phosphododecamolybdate)
6. $\text{H}_6[\text{P}_2\text{W}_{18}\text{O}_{62}] = \text{H}_6[\text{W}_{18}\text{O}_{54}(\text{PO}_4)_2]$
hexahydrogen[dohexacontaoxido(diphosphorusoctadecatungsten)ate], or
hexahydrogen[tetrapentacontaoxidobis(tetraoxidophosphato)octadecatungstate],
or hexahydrogen(diphosphooctadecatungstate)
7. $\text{H}_4[\text{Fe}(\text{CN})_6]$
tetrahydrogen(hexacyanidoferrate)
8. $\text{H}_2[\text{PtCl}_6] \cdot 2\text{H}_2\text{O}$
dihydrogen(hexachloridoplatinate)—water (1/2)
9. HCN
hydrogen(nitridocarbonate)

In Example 1, the two hydrons could be located either on two oxygen atoms on the same phosphorus atom or one on each of the phosphorus atoms. Thus, as already indicated, hydrogen names do not necessarily fully specify the structure.

In the same way, the hydrogen name in Example 9 covers, in principle, two tautomers. This also applies to the common compositional name ‘hydrogen cyanide’. The names ‘hydridonitridocarbon’ (additive nomenclature), ‘methyldyneazane’ (substitutive nomenclature) and ‘formonitrile’ (functional organic nomenclature) all specify the tautomer HCN.

Hydrogen names may also be used for molecular compounds and ions with no tautomerism problems if one wishes to emphasize the conception of the structure as hydrons attached to the anion in question:

Examples:

- | | |
|------------------------------|---------------------------------|
| 10. HMnO_4 | hydrogen(tetraoxidomanganate) |
| 11. H_2MnO_4 | dihydrogen(tetraoxidomanganate) |

12. H_2CrO_4	dihydrogen(tetraoxidochromate)
13. HCrO_4^-	hydrogen(tetraoxidochromate)(1-)
14. $\text{H}_2\text{Cr}_2\text{O}_7$	dihydrogen(heptaoxidodichromate)
15. H_2O_2	dihydrogen(peroxide)
16. HO_2^-	hydrogen(peroxide)(1-)
17. H_2S	dihydrogen(sulfide)
18. H_2NO_3^+	dihydrogen(trioxidonitrate)(1+)

Note the difference from *compositional names* such as ‘hydrogen peroxide’ for H_2O_2 and ‘hydrogen sulfide’ for H_2S (Chapter IR-5) in which (in English) there is a space between the electropositive and electronegative component(s) of the name.

Compositional names of the above type, containing the word ‘hydrogen’, were classified as ‘hydrogen nomenclature’ in the discussion of oxoacids in Section I-9.5 of Ref. 2, and such names were extensively exemplified. However, in order to avoid ambiguity, their general use is not encouraged here. Consider, for example, that the compositional names ‘hydrogen sulfide’ and ‘hydrogen sulfide(2-)’ can both be interpreted as H_2S as well as HS^- . The situation with H_2S is completely analogous to that with Na_2S which may be named sodium sulfide, disodium sulfide, sodium sulfide(2-) and disodium sulfide(2-), except that misinterpretation of the first and third names as denoting NaS^- is improbable. In Ref. 2, the names ‘hydrogensulfide(1-)’ and ‘monohydrogensulfide’ for HS^- were proposed to avoid ambiguity. (However, in some languages there is no space in compositional names so that very delicate distinctions are required anyway.)

The strict definition of *hydrogen names* proposed here is meant to eliminate such confusion by imposing the requirements:

- (i) that ‘hydrogen’ be attached to the rest of the name,
- (ii) that the number of hydrogens *must* be specified by a multiplicative prefix,
- (iii) that the anionic part be placed in enclosing marks, and
- (iv) that the charge of the *total* structure being named is specified.

Hydrogen names constructed in this way cannot be mistaken for other types of name.

The only acceptable exceptions to the above format for hydrogen names are the few particular abbreviated anion names listed in Section IR-8.5.

In a few cases, no confusion can arise, and the distinction between compositional name and hydrogen name is not as important, most notably for the hydrogen halides. Thus, HCl can equally unambiguously be named ‘hydrogen chloride’ (compositional name) and ‘hydrogen(chloride)’ (hydrogen name).

Examples 1, 3–6 and 14 above demonstrate that homo- and heteropolyoxoacids and their partially dehydrated forms can be given hydrogen names once the corresponding anions have been named. Examples 4–6 each feature three alternatives. The first two names are both fully additive for the anion part and correspond to different ways of dissecting the structure into ligands and central atoms. The last names, involving the prefixes ‘silico’ and ‘phospho’, are examples of a common semi-systematic nomenclature which is not recommended for general use because it requires complex conventions in order to be unambiguous.

Rules for naming very complicated homo- and heteropolyoxoanions are given in Chapter II-1 of Ref. 3.

Note that Examples 10–14 above show how one may easily name transition metal compounds that have been named as acids in the past. Names such as permanganic acid, dichromic acid, *etc.*, are not included in the present recommendations because they represent an area where it is difficult to systematize and decide what to include, and where the names are not needed for organic nomenclature, as opposed to the corresponding ‘acid’ names for acids of main group elements.

Finally, note that usage is different from the above in the names of salts and partial esters of organic polyvalent acids, where ‘hydrogen’ is always cited as a separate word just before the anion name, *e.g.* potassium hydrogen phthalate or ethyl hydrogen phthalate.

IR-8.5

ABBREVIATED HYDROGEN NAMES FOR CERTAIN ANIONS

A few common anionic species have names which can be regarded as short forms of hydrogen names formed according to the above method. These names, all in one word without explicit indication of the molecular charge, and without the enclosing marks, are accepted due to their brevity and long usage and because they are not ambiguous. It is strongly recommended that this list be viewed as limiting due to the ambiguities that may arise in many other cases. (See the discussion in Section IR-8.4.)

<i>Anion</i>	<i>Accepted simplified hydrogen name</i>	<i>Hydrogen name</i>
H_2BO_3^-	dihydrogenborate	dihydrogen(trioxidoborate)(1–)
HBO_3^{2-}	hydrogenborate	hydrogen(trioxidoborate)(2–)
HSO_4^-	hydrogensulfate	hydrogen(tetraoxidosulfate)(1–)
HCO_3^-	hydrogencarbonate	hydrogen(trioxidocarbonate)(1–)
H_2PO_4^-	dihydrogenphosphate	dihydrogen(tetraoxidophosphate)(1–)
HPO_4^{2-}	hydrogenphosphate	hydrogen(tetraoxidophosphate)(2–)
HPO_3^-	hydrogenphosphonate	hydrogen(hydridotrioxidophosphate)(1–)
H_2PO_3^-	dihydrogenphosphite	dihydrogen(trioxidophosphate)(1–)
HPO_3^{2-}	hydrogenphosphite	hydrogen(trioxidophosphate)(2–)
HSO_4^-	hydrogensulfate	hydrogen(tetraoxidosulfate)(1–)
HSO_3^-	hydrogensulfite	hydrogen(trioxidosulfate)(1–)

IR-8.6

FUNCTIONAL REPLACEMENT NAMES FOR DERIVATIVES OF OXOACIDS

In functional replacement nomenclature, substitution of =O or –OH groups in parent oxoacids (such as $\text{O} \rightarrow \text{S}$, $\text{O} \rightarrow \text{OO}$, $\text{OH} \rightarrow \text{Cl}$, *etc.*) is indicated by the use of infixes or prefixes as exemplified below (see Ref. 1, Section P–67.1).

<i>Replacement operation</i>	<i>Prefix</i>	<i>Infix</i>
$\text{OH} \rightarrow \text{NH}_2$	amid(o)	amid(o)
$\text{O} \rightarrow \text{OO}$	peroxy	peroxo
$\text{O} \rightarrow \text{S}$	thio	thio
$\text{O} \rightarrow \text{Se}$	seleno	seleno
$\text{O} \rightarrow \text{Te}$	telluro	telluro
$\text{OH} \rightarrow \text{F}$	fluoro	fluorid(o)
$\text{OH} \rightarrow \text{Cl}$	chloro	chlorid(o)
$\text{OH} \rightarrow \text{Br}$	bromo	bromid(o)
$\text{OH} \rightarrow \text{I}$	iodo	iodid(o)
$\text{OH} \rightarrow \text{CN}$	cyano	cyanid(o)

Example 5 in Section IR-8.1 demonstrates the use of the infixes for $\text{OH} \rightarrow \text{Cl}$ and $\text{O} \rightarrow \text{S}$ to arrive at the name ‘arsonochloridothioic *O*-acid’ for the derived parent $\text{HAsCl(OH)S} = [\text{AsClH(OH)S}]$, required for naming the organic derivative:

EtAsCl(OH)S ethylarsonochloridothioic *O*-acid.

Functional replacement names may, of course, be used for the derived parent acids themselves. However, this amounts to introducing an additional system which is not needed in inorganic nomenclature. As mentioned above, additive and substitutive nomenclature can always be used.

Example:

1. $\text{HAsCl(OH)S} = [\text{AsClH(OH)S}]$
chloridohydridohydroxidosulfidoarsenic (additive), or
chloro(hydroxy)- λ^5 -arsanethione (substitutive)

Nevertheless, in Table IR-8.2 several inorganic species are listed which can be regarded as derived from species in Table IR-8.1 by various replacement operations, and for which the common names are in fact derived by the above prefix method (*e.g.* ‘thiosulfuric acid’).

A problem that would arise with the general use of the prefix variant of functional replacement names is illustrated by the thio acids. The names trithiocarbonic acid, tetrathiophosphoric acid, *etc.*, would lead to anion names trithiocarbonate, tetrathiophosphate, *etc.*, which appear to be additive names but are incorrect as such because the ligand prefix is now ‘sulfido’ or ‘sulfanediido’ [thus giving trisulfidocarbonate(2–), tetrasulfidophosphate(3–), *etc.*]. Section P-65.2 of Ref. 1 prescribes the infix-based name carbonotrithioic acid, leading to the anion name carbonotrithioate, which will not be mistaken for an additive name.

A few examples of other functional nomenclature are also included in Table IR-8.2 (*e.g.* phosphoryl chloride, sulfuric diamide). These particular names are well entrenched and can still be used, but this type of nomenclature is not recommended for compounds other than those shown. Again, additive and substitutive names may always be constructed, as exemplified in the Table.

Table IR-8.2 *Acceptable common names, functional replacement names, and systematic (additive) names for some functional replacement derivatives of oxoacids*

This Table gives acceptable common names, functional replacement names (see Section IR-8.6) and systematic (additive) names for compounds related to oxoacids in Table IR-8.1 and certain isomers and corresponding anions. The examples given are derived by formal replacement of an O atom/O atoms, or of an OH group/OH groups, by (an)other atom(s) or group(s).

Formulae are in some cases given in the classical format with the 'acid' (oxygen- or chalcogen-bound) hydrogen atoms listed first (e.g. $\text{H}_2\text{S}_2\text{O}_3$). In most cases formulae are also (or only) written as coordination entities, assembled according to the principles of Chapter IR-7 (e.g. $[\text{H}_2\text{S}_2\text{O}_3]^-$ (e.g. $[\text{SO}(\text{OH})_2\text{S}]^-$).

<i>Formula</i>	<i>Acceptable common name</i>	<i>Functional replacement name</i>	<i>Systematic (additive) name</i>
$\text{HNO}_4 = [\text{NO}_2(\text{OOH})]$	peroxynitric acid ^a	peroxynitric acid ^a	(dioxidano)dioxidonitrogen
$\text{NO}_4^- = [\text{NO}_2(\text{OO})]^-$	peroxynitrate ^a	peroxynitrate ^a	dioxidoperoxidonitrate(1-)
$[\text{NO}(\text{OOH})]$	peroxynitrous acid ^a	peroxynitrous acid ^a	(dioxidano)oxidonitrogen
$[\text{NO}(\text{OO})]^-$	peroxynitrite ^a	peroxynitrite ^a	oxidoperoxidonitrate(1-)
$\text{NO}_2\text{NH}_2 = \text{N}(\text{NH}_2)\text{O}_2$	nitramide	nitric amide	amidodioxidonitrogen, or dihydrido-1 κ^2 H-dioxido-2 κ^2 O-dinitrogen(N—N)
$\text{H}_3\text{PO}_5 = [\text{PO}(\text{OH})_2(\text{OOH})]$	peroxyphosphoric acid ^a	phosphoroperoxoic acid	(dioxidano)dihydrodioxidoposphorus
$[\text{PO}_5]^{3-} = [\text{PO}_3(\text{OO})]^{3-}$	peroxyphosphate ^a	phosphoroperoxoate	trioxidoperoxidophosphate(3-)
$[\text{PCL}_3\text{O}]$	phosphoryl trichloride, or phosphorus trichloride oxide	phosphoryl trichloride	trichloridooxidophosphorus
$\text{H}_4\text{P}_2\text{O}_8 = [(\text{HO})_2\text{P}(\text{O})(\text{OOP}(\text{O})(\text{OH})_2)]$	peroxydiphosphoric acid ^a	2-peroxydiphosphoric acid	μ -peroxido-1 $\kappa\text{O}, 2\kappa\text{O}'$ -bis(dihydrodioxidoposphorus)
$[\text{P}_2\text{O}_8]^{4-} = [\text{O}_3\text{POOP}_3]^{4-}$	peroxydiphosphate ^a	2-peroxydiphosphate	μ -peroxido-1 $\kappa\text{O}, 2\kappa\text{O}'$ -bis(trioxidophosphate)(4-)
$\text{H}_2\text{SO}_5 = [\text{SO}_2(\text{OH})(\text{OOH})]$	peroxysulfuric acid ^a	sulfuroperoxoic acid	(dioxidano)hydroxidodioxidosulfur
$[\text{SO}_5]^{2-} = [\text{SO}_3(\text{OO})]^{2-}$	peroxysulfate ^a	sulfuroperoxoate	trioxidoperoxidosulfate(2-)
$\text{H}_2\text{S}_2\text{O}_8 = [(\text{HO})\text{S}(\text{O})_2\text{OOS}(\text{O})_2(\text{OH})]$	peroxydisulfuric acid ^a	2-peroxydisulfuric acid	μ -peroxido-1 $\kappa\text{O}, 2\kappa\text{O}'$ -bis(hydroxidodioxidosulfur)
$[\text{S}_2\text{O}_8]^{2-} = [\text{O}_3\text{SOOSO}_3]^{2-}$	peroxydisulfate ^a	2-peroxydisulfate	μ -peroxido-1 $\kappa\text{O}, 2\kappa\text{O}'$ -bis(trioxidosulfate)(2-)
$\text{H}_2\text{S}_2\text{O}_3 = [\text{SO}(\text{OH})_2\text{S}]$	thiosulfuric acid	sulfurothioic O-acid	dihydrodioxidosulfidosulfur
$\text{H}_2\text{S}_2\text{O}_3 = [\text{SO}_2(\text{OH})(\text{SH})]$	thiosulfuric acid	sulfurothioic S-acid	hydroxidodioxidosulfanidosulfur
$\text{S}_2\text{O}_3^{2-} = [\text{SO}_3\text{S}]^{2-}$	thiosulfate	sulfurothioate	trioxidosulfidosulfate(2-)
$\text{H}_2\text{S}_2\text{O}_2 = [\text{S}(\text{OH})_2\text{S}]$	thiosulfurous acid	sulfurothious O-acid	dihydrodioxidosulfidosulfur
$\text{H}_2\text{S}_2\text{O}_2 = [\text{SO}(\text{OH})(\text{SH})]$	thiosulfurous acid	sulfurothious S-acid	hydroxidooxidosulfanidosulfur

(continued)

Table IR-8.2 *Continued*

<i>Formula</i>	<i>Acceptable common name</i>	<i>Functional replacement name</i>	<i>Systematic (additive) name</i>
$[\text{SO}_2\text{S}]^{2-}$	thiosulfite	sulfurothioite	dioxidosulfidosulfate(2-)
$\text{SO}_2\text{Cl}_2 = [\text{SCl}_2\text{O}_2]$	sulfuryl dichloride, or sulfur dichloride dioxide	sulfuryl dichloride	dichloridodioxidosulfur
$\text{SOCl}_2 = [\text{SCl}_2\text{O}]$	thionyl dichloride, or sulfur dichloride oxide	sulfurous dichloride	dichloridooxidosulfur
$[\text{S}(\text{NH}_2)\text{O}_2(\text{OH})]$	sulfamic acid	sulfuramidic acid	amidohydroxidodioxidosulfur
$[\text{S}(\text{NH}_2)_2\text{O}_2]$	sulfuric diamide	sulfuric diamide	diamidodioxidosulfur
$\text{HSCN} = [\text{C}(\text{N})(\text{SH})]$	thiocyanic acid		nitridosulfanidocarbon
$\text{HNCS} = [\text{C}(\text{NH})\text{S}]$	isothiocyanic acid		imidosulfidocarbon
SCN^-	thiocyanate		nitridosulfidocarbonate(1-)

^a These names were given with the prefix 'peroxo' rather than 'peroxy' in Ref. 4 (Rule 5.22). However, in Ref. 2 names with the prefix 'peroxo' were dismissed, with no reason given, and no other prefixes were provided instead. The names with the prefix 'peroxy' continue to be in frequent use. Furthermore, the general rule in functional replacement nomenclature (Ref. 1, Sec. P-15.5) is that the replacement *prefix* for the replacement $-\text{O}- \rightarrow -\text{OO}-$ is, indeed, 'peroxy' (as opposed to the *infix* for this replacement, which is 'peroxo'). In view of this, the names with the prefix 'peroxy' are listed here. For most mononuclear oxoacids, the present rules in Ref. 1 (Section P-67.1) prescribe using the infix method for systematic names; in those cases the resulting names are given in the second column here. The prefix method is used for nitric and nitrous acids and dinuclear oxoacids, as also seen here.

IR-8.7

REFERENCES

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